# ASH-LR: Adaptive and Numerically Stable Optimization for Ill-Conditioned Logistic Regression

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**Abstract**

Fitting $\ell_2$-regularized logistic regression is slow for first-order methods when features are correlated or differently scaled, and naive evaluation of the loss can overflow. We study Adaptive Stable Hybrid Logistic Regression (ASH-LR), which combines overflow-free evaluation of the loss, an initial phase of gradient steps, and a damped inexact Newton phase in which $(H + \mu I)p = -g$ is solved by conjugate gradients using Hessian-vector products, with $\mu = \sqrt{\theta\|g\|}$ and $\theta$ adapted by a ratio test. Following a pre-registered plan, we compare ASH-LR with gradient descent, Newton's method, Newton–CG and L-BFGS on eight regimes (synthetic designs with condition numbers up to $10^6$, badly scaled features, and the Spambase data set), ten seeds each, and complement the comparison with robustness, ablation, single-precision and stability experiments. ASH-LR converged on all problems, whereas L-BFGS failed on the larger and badly scaled ones, and stable evaluation was essential from distant starting points. The pre-registered hypothesis of a 20% reduction in work relative to the best matrix-free baseline was not supported: ASH-LR needed 18–127% more data passes than Newton–CG in four of five ill-conditioned regimes and was statistically on a par in the fifth, and direct Newton was fastest for all dimensions studied ($d \leq 500$). The ablation attributes the gap to the gradient phase and, on badly scaled data, to the accuracy of the unpreconditioned inner solver rather than to damping.

*Keywords:* logistic regression, ill-conditioning, inexact Newton method, conjugate gradient, regularized Newton, numerical stability

## 1. Introduction

Logistic regression remains one of the most widely used models for binary classification, and fitting it is a smooth, strongly convex optimization problem when an $\ell_2$ penalty is added. In practice the problem is often *ill-conditioned*: features are strongly correlated, measured on very different scales, or both, so that the Hessian of the objective has eigenvalues spread over many orders of magnitude. First-order methods then converge slowly, because their step size is limited by the largest curvature while progress along low-curvature directions is proportional to the smallest one [1, 2]. Newton-type methods are insensitive to this spread in theory, but they require second-order information, can take poor steps far from the solution, and, in their textbook form, solve a dense linear system whose cost grows as $O(nd^2 + d^3)$.

A second, more mundane difficulty is numerical. Evaluating $\log(1+e^{-z})$ and the sigmoid $1/(1+e^{-z})$ directly overflows once $|z|$ exceeds roughly 709 in double precision and 88 in single precision. Such margins arise when features are unscaled, when the starting point is far from the solution,

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or when the data are nearly separable, and an overflowing loss silently breaks line searches and acceptance tests [3, 4].

This report studies *Adaptive Stable Hybrid Logistic Regression* (ASH-LR), a method that combines (i) overflow-free evaluation of the loss and its derivatives, (ii) an initial phase of inexpensive gradient steps followed by a one-way switch to second-order steps, and (iii) a damped inexact Newton phase in which the system $(H + \mu I)p = -g$ is solved by conjugate gradients (CG) using Hessian-vector products only, and the damping $\mu$ is adapted by a ratio test that compares the actual with the predicted decrease. In the final version the damping follows the gradient-regularized rule $\mu_k = \sqrt{\theta_k \|g_k\|}$, which is motivated by recent regularized Newton methods [5, 6] and vanishes automatically near the solution.

The project went through two stages. A preliminary version (Milestone 2) used direct linear solves and line-search-based damping; it converged on all test problems but needed more iterations and more time than plain Newton. Exploratory runs then showed that the halving damping rule kept steps over-damped for many iterations. The final method replaces it, and the final evaluation was fixed in a pre-registered plan, committed to the code repository before the final experiments were run, to avoid selecting favourable test cases after the fact.

*Research question.* Can combining gradient steps with a damped, matrix-free Newton–CG phase improve the robustness and the work-to-accuracy of ill-conditioned logistic regression, compared with established matrix-free methods (Newton–CG and L-BFGS)? The pre-registered success criterion (H1) is a reduction of at least 20% in the median number of data passes relative to the best converging matrix-free baseline in at least two of five predeclared ill-conditioned regimes.

*Contributions.*

- A complete, documented and tested implementation of ASH-LR and five baselines with a common interface, an exact work counter in units of data passes, and a generator of synthetic designs with an exactly prescribed condition number.
- A pre-registered evaluation on eight regimes (six synthetic, two based on the Spambase data set) with ten seeds each, complemented by robustness, ablation, single-precision and stability experiments.
- An honest assessment: we report where ASH-LR helps, where it does not, and why direct Newton remains the method of choice for small dense problems.

The rest of the report is organized as follows. Section 2 reviews related work, Section 3 formulates the problem and the algorithm, Section 4 describes the experimental setup, Section 5 presents and discusses the results, and Section 6 concludes.

## 2. Related work

*Second-order methods for logistic regression.* Galli and Lin [7] carried out a systematic study of truncated Newton methods for linear classification, including $\ell_2$-regularized logistic regression. They compared line-search and trust-region globalizations and inner CG stopping rules within the LIBLINEAR framework and found that well-designed truncated Newton methods are robust and efficient across many data sets; Newton–CG with a line search, one of our baselines, belongs to this family. Berahas, Bollapragada and Nocedal [2] compared Newton-Sketch and subsampled Newton methods on logistic regression and showed that inexact Newton-CG steps with Hessian-vector products are competitive with first-order methods on ill-conditioned problems. For quasi-Newton methods,

Berahas et al. [8] proposed sampled L-BFGS variants for machine learning; we use the classical deterministic L-BFGS as the second matrix-free baseline. Wright and Recht [1] give a modern treatment of the line searches, conjugate gradients and Newton-type methods used throughout this report.

*Inexact Newton–CG with guarantees.* Royer, O'Neill and Wright [9] designed a Newton–CG method whose inner CG iterations are capped and safeguarded, and proved worst-case complexity bounds for smooth problems. Yao et al. [10] extended such guarantees to Newton–CG with inexact Hessian information and relaxed inner tolerances, and Lim and Roosta [11] established complexity guarantees for the related Newton-MR method under inexact Hessians. These works justify solving the Newton system only approximately, with a relative residual tolerance (forcing term) that decreases with the gradient norm, as ASH-LR does.

*Regularized and damped Newton methods.* Adding $\mu I$ to the Hessian is the classical Levenberg–Marquardt idea; recent work ties $\mu$ to the gradient norm. Mishchenko [5] proved that $\mu_k = \sqrt{H\|g_k\|}$, with $H$ the Lipschitz constant of the Hessian, gives a global $O(1/k^2)$ rate together with local superlinear convergence without line search, at the cost of one linear solve per iteration. Doikov and Nesterov [12] generalized gradient regularization to Bregman distances, and Doikov, Mishchenko and Nesterov [6] proposed a *super-universal* method that adapts the regularization constant and the power of the gradient norm automatically and evaluated it on logistic regression. Hanzely et al. [13] showed that a damped Newton method with an explicit step size achieves global $O(1/k^2)$ and local quadratic rates for self-concordant-type functions. ASH-LR adopts the form $\mu_k = \sqrt{\theta_k\|g_k\|}$ from this line of work and adapts $\theta_k$ by a ratio test. The guarantees of [5, 6, 12] rely on their exact update rules and exact linear solves, so they do not transfer directly to ASH-LR.

*Numerical stability and precision.* Blanchard, Higham and Higham [3] analysed the rounding errors of log-sum-exp and softmax and showed that the shifted formulation is accurate; the logistic loss is the two-term case, $\log(1 + e^{-z}) = \mathrm{lse}(0, -z)$. Wiemann, Kneib and Hambuckers [4] use the softplus function to build link functions for generalized linear models and rely on the same stable identity. Higham and Mary [14] survey mixed-precision linear algebra and explain when single precision is safe; their analysis of conditioning versus unit roundoff underlies our float32 experiment. Randomized Nyström preconditioning [15] is one recent way to make CG robust to ill-conditioning and is relevant to the bottleneck we identify.

*Streaming second-order methods and benchmarking.* Godichon-Baggioni and Werge [16] develop inversion-free stochastic Newton methods with $O(dN)$ cost for ill-conditioned streaming problems, illustrating both the benefit and the expense of curvature information; our setting is deterministic batch optimization. Finally, Benchopt [17] argues for transparent, reproducible optimization benchmarks with fixed protocols, which motivated the pre-registered design of our evaluation.

## 3. Problem formulation and method

### *3.1. Objective*

Given observations $(x_i, y_i)$, $i = 1, \ldots, n$, with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, we minimize

$$F(w) = \frac{1}{n} \sum_{i=1}^{n} \log\bigl(1 + e^{-z_i}\bigr) + \frac{\lambda}{2}\|w\|_2^2, \qquad z_i = y_i\, x_i^\top w, \tag{1}$$

with $\lambda = 10^{-6}$ unless stated otherwise and no separate intercept. Writing $X \in \mathbb{R}^{n\times d}$ for the design matrix and $\sigma(t) = 1/(1+e^{-t})$,

$$\nabla F(w) = -\frac{1}{n} X^\top \big(y \odot \sigma(-z)\big) + \lambda w, \qquad \nabla^2 F(w) = \frac{1}{n} X^\top D X + \lambda I, \quad D = \operatorname{diag}\big(\sigma(z_i)(1-\sigma(z_i))\big). \tag{2}$$

Since $\lambda > 0$, $F$ is $\lambda$-strongly convex and has a unique minimizer $w^*$. The weights $D_{ii} \le 1/4$ vanish for large $|z_i|$, so the Hessian inherits and can amplify the ill-conditioning of $X^\top X/n$: its smallest eigenvalue can be as small as $\lambda$.

### *3.2. Stable evaluation*

The loss term is evaluated as $\log(1+e^{-z}) = \texttt{logaddexp}(0, -z)$, which computes $\max(0,-z) + \log(1+e^{-|z|})$ and never exponentiates a positive number. The sigmoid uses $\sigma(t) = 1/(1+e^{-t})$ for $t \ge 0$ and $e^t/(1+e^t)$ for $t < 0$. Both forms are exact reformulations, so they change nothing when the naive formulas do not overflow. For the ablation in Section 5 we also implement the naive formulas $\log(1+e^{-z})$ and $1/(1+e^{-z})$.

### *3.3. Search directions and line search*

Let $g_k = \nabla F(w_k)$ and $H_k = \nabla^2 F(w_k)$. The methods compared use

$$\begin{aligned} \text{GD:}\quad & p_k = -g_k, & \text{Newton:}\quad & H_k p_k = -g_k \text{ (dense solve)}, \\ \text{Newton–CG:}\quad & H_k p_k \approx -g_k \text{ (CG)}, & \text{ASH-LR (phase 2):}\quad & (H_k + \mu_k I) p_k \approx -g_k \text{ (CG)}. \end{aligned}$$

L-BFGS uses the two-loop recursion with memory $m = 10$ and skips curvature pairs with $s^\top y \le 10^{-10}\|s\|\|y\|$. Except for GD with fixed step $1/L$, $L = \|X\|_2^2/(4n) + \lambda$, all line searches use Armijo backtracking [1]: starting from $\alpha = 1$, the step is halved until $F(w_k + \alpha p_k) \le F(w_k) + c\,\alpha\, g_k^\top p_k$ with $c = 10^{-4}$, for at most 40 trials.

### *3.4. Inexact solves with Hessian-vector products*

The product $(H_k + \mu I)v = \frac{1}{n} X^\top \big(D_k (Xv)\big) + (\lambda + \mu) v$ costs two passes over the data and never forms $H_k$. CG is run on $(H_k + \mu_k I)p = -g_k$ from $p = 0$ and stopped when the residual $r = -g_k - (H_k + \mu_k I)p$ satisfies

$$\|r\| \le \eta_k \|g_k\|, \qquad \eta_k = \min\big(0.5, \sqrt{\|g_k\|}\big), \tag{3}$$

or after $d$ iterations. A forcing term that decreases with $\|g_k\|$ preserves the fast local convergence of Newton's method while keeping early inner solves cheap [9, 10]. Because $H_k + \mu_k I$ is positive definite, every CG iterate is a descent direction [1].

### *3.5. Switching rule*

ASH-LR starts with Armijo gradient steps and switches, once and for all, to the damped Newton phase as soon as either (i) $\|g_k\| \le 0.1\,\|g_0\|$, or (ii) at least five gradient steps have been taken and the last step achieved a relative decrease $(F(w_{k-1}) - F(w_k))/\max(1, |F(w_k)|) \le 10^{-3}$. The idea is that cheap gradient steps make fast progress while far from the solution and while the curvature model is unreliable, and that Newton steps take over when gradient steps stall on the low-curvature directions.

**Algorithm 1** ASH-LR

**Require:** $w_0$, tolerance $\varepsilon$, $\mu_0 = 10^{-3}$

phase $\leftarrow$ GD; $k \leftarrow 0$
**while** $\|g_k\| > \varepsilon$ and $k < k_{\max}$ **do**
  **if** phase = GD **then**
    $p \leftarrow -g_k$; $\alpha \leftarrow \text{Armijo}(w_k, p)$; $w_{k+1} \leftarrow w_k + \alpha p$
    **if** $\|g_{k+1}\| \le 0.1\|g_0\|$ **or** (GD steps $\ge 5$ **and** rel. decrease $\le 10^{-3}$) **then**
      phase $\leftarrow$ Newton; $\theta \leftarrow \mu_0^2/\|g_{k+1}\|$
    **end if**
  **else**
    **for** $j = 1, \dots, 30$ **do**
      $\mu \leftarrow \text{clip}(\sqrt{\theta\|g_k\|})$; solve $(H_k + \mu I)p = -g_k$ by CG to tolerance (3)
      $\rho \leftarrow (F(w_k) - F(w_k + p))/\text{pred}(p)$
      $\theta \leftarrow 4\theta$ if $\rho < 0.25$;  $\theta \leftarrow \theta/2$ if $\rho > 0.75$
      **if** $\rho \ge 10^{-4}$ **then**
        $w_{k+1} \leftarrow w_k + p$; **break**
      **end if**
    **end for**
    **if** no step accepted: $w_{k+1} \leftarrow w_k + \text{Armijo}(w_k, p)\, p$
  **end if**
  $k \leftarrow k + 1$
**end while**

### *3.6. Damping by a ratio test*

For a trial direction $p$ the quadratic model predicts the decrease

$$\text{pred}(p) = -\big(g_k^\top p + \tfrac{1}{2} p^\top H_k p\big). \tag{4}$$

With an inexact solve, $(H_k+\mu_k I)p = -g_k - r$, so $p^\top H_k p = -p^\top g_k - p^\top r - \mu_k\|p\|^2$ is available from CG quantities without an extra Hessian-vector product. The ratio $\rho = \big(F(w_k) - F(w_k + p)\big)/\text{pred}(p)$ measures how well the model predicted the actual decrease. The step is accepted with $\alpha = 1$ if $\rho \ge 10^{-4}$; otherwise it is rejected and the direction is recomputed with larger damping. The damping parameter follows the gradient-regularized rule

$$\mu_k = \text{clip}\big(\sqrt{\theta_k\|g_k\|},\, 10^{-12},\, 10^{8}\big), \tag{5}$$

and the ratio test adapts $\theta_k$: $\theta \leftarrow 4\theta$ if $\rho < 0.25$, $\theta \leftarrow \theta/2$ if $\rho > 0.75$, unchanged otherwise. At the switch $\theta$ is initialized so that $\mu = 10^{-3}$. Because $\mu_k \to 0$ as $\|g_k\| \to 0$, the method becomes an undamped inexact Newton method near the solution, whereas in the Milestone 2 rule $\mu$ was only halved after each full step and could remain large for many iterations. As a safeguard, after 30 consecutive rejections the last direction is used with Armijo backtracking.

Algorithm 1 summarizes the method. Every accepted step decreases $F$: gradient and fallback steps satisfy the Armijo condition, and ratio-test steps satisfy $F(w_k) - F(w_{k+1}) \ge 10^{-4}\,\text{pred}(p) > 0$.

### *3.7. Cost model*

We measure work in *data passes*, one pass being a product with $X$ or $X^\top$ ($\approx nd$ multiply-adds). A function value costs one pass, a gradient at a point whose margins are cached one more pass, a Hessian-vector product two passes, forming $X^\top DX$ costs $d$ passes, and a dense solve $d^2/(3n)$ passes.

Table 1: Regimes of the study. $\kappa$ is the condition number of $X^\top X/n$ (synthetic) and $\kappa(H_0)$ the median condition number of the Hessian at $w_0 = 0$ over the ten seeds (training data).

| ID | Data | Description | $n_{\text{train}}$ | $d$ | $\kappa(H_0)$ |
|---|---|---|---|---|---|
| R1 | synthetic | $\kappa = 10^1$ | 750 | 10 | 9.8e+00 |
| R2 | synthetic | $\kappa = 10^3$ | 750 | 10 | 1.0e+03 |
| R3 | synthetic | $\kappa = 10^6$ | 750 | 10 | 2.0e+05 |
| R4 | synthetic | $\kappa = 10^4$ | 3,000 | 200 | 9.9e+03 |
| R5 | synthetic | $\kappa = 10^6$ | 4,500 | 500 | 2.0e+05 |
| R6 | synthetic | $\kappa = 10^2$, scaled columns $[10^{-2}, 10^2]$ | 1,500 | 50 | 3.2e+08 |
| R7 | Spambase | standardized | 3,451 | 57 | 1.5e+03 |
| R8 | Spambase | raw features | 3,451 | 57 | 9.5e+08 |

Passes are machine-independent and are the primary efficiency measure; wall time is reported as well, because dense matrix–matrix products run much closer to peak speed than the matrix–vector products used by CG, so equal passes do not mean equal time.

## 4. Experimental setup

### *4.1. Data and regimes*

*Synthetic designs.* We generate $X = Z\,\mathrm{diag}(\sqrt{e})\,V^\top$, where $Z \in \mathbb{R}^{n\times d}$ has orthonormal columns scaled so that $Z^\top Z/n = I$, $V$ is a Haar-random orthogonal matrix and the eigenvalues $e_j$ are log-spaced between 1 and $1/\kappa$. Hence $X^\top X/n = V\mathrm{diag}(e)V^\top$ has condition number exactly $\kappa$, and designs with the same seed but different $\kappa$ share $Z$ and $V$. Optionally, columns are multiplied by log-spaced factors in $[10^{-2}, 10^2]$ to create differently scaled features. A true weight vector is drawn at random and rescaled so that the logits $Xw^\star$ have standard deviation 1.5; labels are drawn as $y_i = +1$ with probability $\sigma(x_i^\top w^\star)$.

*Spambase.* The Spambase data set [18] contains 4,601 e-mails described by 57 numeric features (word and character frequencies and capital-run statistics) with 1,813 spam messages. The features range over several orders of magnitude: the word and character frequencies are percentages, whereas `capital_run_length_total` reaches 15,841. Hence the raw data are a natural real-world instance of bad scaling. The file contains 391 exact duplicate rows. They are retained, since they are part of the empirical objective, but test accuracies are interpreted with this caveat.

All data sets use a stratified 75/25 train/test split; standardization (R7) uses training statistics only. Table 1 lists the eight regimes. R3–R6 and R8 are the five primary ill-conditioned regimes of hypothesis H1.

### *4.2. Experiments*

The protocol was fixed in a pre-registered plan (file `EXPERIMENT_PLAN.md` of the repository) before the final runs; exploratory runs that preceded it used a different seed and are disclosed there.

**E1 (main)** All eight regimes, ten seeds $(1000, \ldots, 1009$; data generation for R1–R6, split for R7–R8), $w_0 = 0$, $\lambda = 10^{-6}$, six methods: GD with Armijo, GD with fixed step $1/L$, Newton (dense solve), Newton–CG, L-BFGS and ASH-LR.

**E2 (robustness)** R2, R5, R8 with three seeds, $\lambda \in \{10^{-6}, 10^{-4}, 10^{-2}\}$ and three starting points: $w_0 = 0$, and random directions rescaled so that the logits have standard deviation 5 (*near*) or 50 (*far*). The relative stopping rule $\|g\|/\max(1, \|g_0\|) \le 10^{-6}$ is used, since $\|g_0\|$ can exceed one for random starts.

**E3 (ablation)** R2, R5, R6, R8 with ten seeds and six variants of ASH-LR: (a) the Milestone 2 version (dense solve, line-search damping, $\mu$ adapted directly); (b) CG with line-search damping; (c) CG with the ratio test but $\mu$ adapted directly; (d) the full method; (e) the full method without the gradient phase; (f) dense solve with the ratio test and gradient-regularized $\mu$.

**E4 (single precision)** R2, R7, R8 with three seeds; Newton, Newton–CG and ASH-LR in float32 and float64 with tolerance $10^{-4}$. The objective of each solution is re-evaluated in float64.

**E5 (stability)** R2 and R8 with three seeds from an extreme starting point (logit standard deviation $10^3$), with the stable and the naive formulas, in both precisions.

### *4.3. Stopping, metrics and timing*

A run stops when $\|g\|_2 \le 10^{-6}$, after 1,000 iterations, or when a line search fails. On every problem a reference value $F^*$ is computed by Newton's method with tolerance $10^{-10}$. We report iterations, data passes, wall time, the relative gap $(F - F^*)/\max(1, |F^*|)$, test accuracy, total CG iterations and rejected steps, and the passes and time needed to first reach relative gaps of $10^{-4}$ and $10^{-8}$. For timing, each method is run once untimed (warm-up) and then three times, with the order of methods randomized in each round; the median is reported. All other metrics are deterministic and identical across repetitions. For H1 and for pairwise comparisons we compute, per regime, the median over seeds of the ratio of ASH-LR's passes (or time) to the baseline's and a two-sided Wilcoxon signed-rank test over the ten seeds.

### *4.4. Implementation and environment*

The code is written in Python 3.13 with NumPy 2.5 (OpenBLAS) and pandas; SciPy is used only for the Wilcoxon test. All methods share the same objective, line search, stopping rule and work counter. Correctness is checked by 26 unit tests, including centered finite-difference checks of the gradient and Hessian, a comparison of Hessian-vector products with the dense Hessian, finite outputs at logits of $\pm 10^4$, monotone decrease of the objective for every method, and determinism. Experiments ran on a single virtual machine with two cores of an Intel Xeon processor at 2.10 GHz and 8 GB of memory, one experiment at a time.

## 5. Results and discussion

### *5.1. Main comparison (E1)*

Table 2 reports, for every regime and method, the number of seeds on which the gradient tolerance was reached, together with median iterations, data passes, wall time and test accuracy. Three observations stand out.

*Reliability.* ASH-LR, Newton and Newton–CG converged on all 80 problems. Both gradient methods failed to reach the tolerance within 1,000 iterations on every ill-conditioned regime (R2–R8), and L-BFGS failed on all seeds of R5, R6 and R8, i.e. in the larger and the badly scaled problems. On raw Spambase (R8), the unconverged gradient methods stopped at test accuracies of 0.44–0.56, against 0.91 for the converged solutions, which shows that non-convergence is not harmless. Converged methods reached the same objective value up to a relative gap of about $10^{-7}$, and their test accuracies differed by at most 0.002 on the same seed.

*Direct Newton is the fastest method.* With $d \le 500$ dense features, forming $X^\top DX$ is a single BLAS-3 operation, and Newton needed 4–13 iterations everywhere. In wall time it was faster than every other method in every regime; in R4–R8 the next fastest converged method was 2 to 240 times slower. In data passes it was also the cheapest method except in R1 and R7, where L-BFGS was

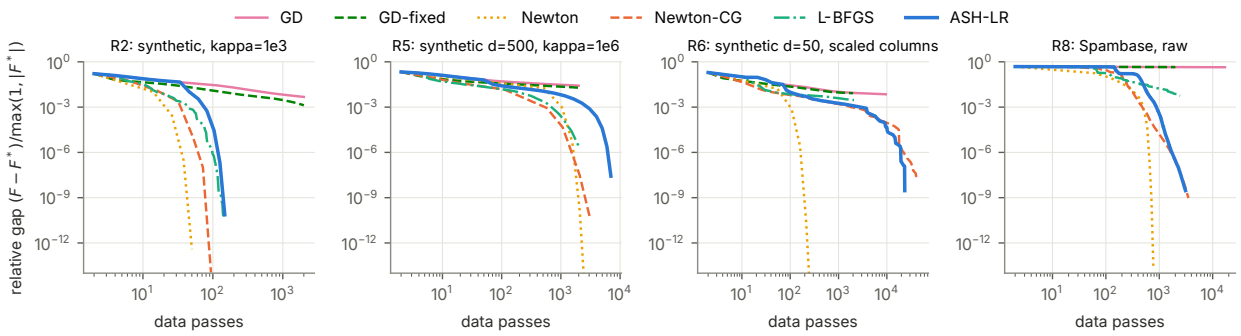



Figure 1: Relative objective gap versus data passes (seed 1000, log–log scale) in four regimes. Direct Newton (dotted) reaches machine precision in a few hundred passes in R6 and R8, where the matrix-free methods need tens of thousands.

cheaper, and in R4, where forming the $200 \times 200$ Hessian costs $d$ passes per iteration and Newton–CG used fewer passes.

*Among matrix-free methods, Newton–CG is more efficient than ASH-LR.* ASH-LR required more passes than Newton–CG in every regime except R6, where the two are statistically indistinguishable (Table 3). It was more efficient than L-BFGS where L-BFGS struggled but converged (R3: median passes ratio 0.71 and five times faster), and it converged where L-BFGS did not (R5, R6, R8).

Figure 1 shows the relative objective gap against data passes for one seed in four representative regimes. The curves make the cost structure of ASH-LR visible: during the gradient phase (the first 15–40 passes) it tracks GD, and only after the switch does it descend at the rate of a Newton-type method. In R2 and R5 the gradient phase is pure overhead. In R6 and R8 ASH-LR and Newton–CG follow almost the same path, with both needing hundreds of CG iterations per Newton step.

### *5.2. Pre-registered hypothesis H1*

Table 3 gives, for the five primary regimes, the per-seed ratios of ASH-LR's passes and times to those of each baseline that converged on all seeds. The best converging matrix-free baseline was Newton–CG in all five regimes, and the change in median passes of ASH-LR relative to it was +111% (R3), +126% (R4), +127% (R5), +17% (R6) and +18% (R8). ASH-LR therefore did not achieve the required reduction of 20% in any regime, and **H1 is not supported**. In R6 the median per-seed ratio is 0.93 and the Wilcoxon test does not detect a difference ($p = 0.92$): ASH-LR was cheaper on some seeds and more expensive on others, and the ratio of the medians (1.17) and the median of the ratios (0.93) point in different directions, which itself indicates that the two methods are on a par there.

Figure 2 extends the comparison to all regimes. ASH-LR uses between about 1.2 and 2.3 times the passes of Newton–CG, except in R6, and the ratio to direct Newton grows to two orders of magnitude in the badly scaled synthetic regime.

### *5.3. Ablation (E3)*

Table 4 isolates the components of ASH-LR, and Figure 3 displays the median passes.

*The gradient phase is the main overhead.* Removing it (variant e) reduced the median passes in R2, R5 and R6 (by 35%, 19% and 27%) and, in R2 and R5, cut the number of iterations by roughly 70%. Only on raw Spambase did it help slightly (3,121 vs. 3,193 passes). Because Newton's method on the logistic loss is well behaved from $w_0 = 0$, as also observed for truncated Newton methods on linear classification [7], the cautious start that motivated the gradient phase is rarely needed. This finding is post hoc: the final method was frozen with the gradient phase, as specified.

Table 2: E1: main comparison, $w_0 = 0$, $\lambda = 10^{-6}$, tolerance $\|g\| \le 10^{-6}$. Conv. = seeds converged out of ten; the other columns are medians over the ten seeds. Passes are products with $X$ or $X^\top$.

| Regime | Method | Conv. | Iter. | Passes | Time (ms) | Test acc. |
|---|---|---|---|---|---|---|
| R1 | GD | 10/10 | 566 | 1,135 | 27.1 | 0.736 |
| | GD-fixed | 10/10 | 136 | 273 | 6.2 | 0.736 |
| | Newton | 10/10 | 4 | 50 | 0.6 | 0.736 |
| | Newton-CG | 10/10 | 5 | 50 | 0.7 | 0.736 |
| | L-BFGS | 10/10 | 16 | 34 | 1.4 | 0.736 |
| | **ASH-LR** | 10/10 | 27 | 94 | 1.9 | 0.736 |
| R2 | GD | 0/10 | 1000 | 2,002 | 49.4 | 0.746 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 44.9 | 0.754 |
| | Newton | 10/10 | 4 | 50 | 0.6 | 0.746 |
| | Newton-CG | 10/10 | 6 | 92 | 1.0 | 0.746 |
| | L-BFGS | 10/10 | 60 | 128 | 5.2 | 0.746 |
| | **ASH-LR** | 10/10 | 23 | 140 | 2.2 | 0.746 |
| R3 | GD | 0/10 | 1000 | 2,002 | 59.8 | 0.738 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 54.9 | 0.736 |
| | Newton | 10/10 | 4 | 50 | 0.8 | 0.742 |
| | Newton-CG | 10/10 | 8 | 126 | 1.8 | 0.742 |
| | L-BFGS | 10/10 | 192 | 416 | 25.5 | 0.742 |
| | **ASH-LR** | 10/10 | 28 | 266 | 4.1 | 0.742 |
| R4 | GD | 0/10 | 1000 | 2,002 | 554 | 0.722 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 534 | 0.725 |
| | Newton | 10/10 | 5 | 1,034 | 36.9 | 0.721 |
| | Newton-CG | 10/10 | 6 | 753 | 152 | 0.721 |
| | L-BFGS | 10/10 | 467 | 952 | 299 | 0.720 |
| | **ASH-LR** | 10/10 | 34 | 1,704 | 363 | 0.721 |
| R5 | GD | 0/10 | 1000 | 2,002 | 1,557 | 0.717 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 1,693 | 0.716 |
| | Newton | 10/10 | 5 | 2,605 | 265 | 0.705 |
| | Newton-CG | 10/10 | 7 | 3,082 | 1,971 | 0.705 |
| | L-BFGS | 0/10 | 1000 | 2,035 | 1,652 | 0.705 |
| | **ASH-LR** | 10/10 | 37 | 6,990 | 4,645 | 0.705 |
| R6 | GD | 0/10 | 1000 | 9,462 | 496 | 0.727 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 89.1 | 0.727 |
| | Newton | 10/10 | 5 | 265 | 3.4 | 0.728 |
| | Newton-CG | 10/10 | 492 | 46,162 | 828 | 0.728 |
| | L-BFGS | 0/10 | 1000 | 2,086 | 153 | 0.725 |
| | **ASH-LR** | 10/10 | 598 | 54,120 | 901 | 0.728 |
| R7 | GD | 0/10 | 1000 | 2,002 | 236 | 0.920 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 239 | 0.920 |
| | Newton | 10/10 | 10 | 595 | 16.0 | 0.921 |
| | Newton-CG | 10/10 | 12 | 600 | 34.4 | 0.921 |
| | L-BFGS | 10/10 | 142 | 294 | 49.7 | 0.921 |
| | **ASH-LR** | 10/10 | 20 | 740 | 45.3 | 0.921 |
| R8 | GD | 0/10 | 1000 | 16,906 | 2,379 | 0.564 |
| | GD-fixed | 0/10 | 1000 | 2,002 | 250 | 0.441 |
| | Newton | 10/10 | 13 | 773 | 19.3 | 0.909 |
| | Newton-CG | 10/10 | 62 | 2,647 | 131 | 0.909 |
| | L-BFGS | 0/10 | 1000 | 2,449 | 372 | 0.916 |
| | **ASH-LR** | 10/10 | 76 | 3,122 | 180 | 0.909 |

Table 3: H1: per-seed ratios ASH-LR / baseline (median and interquartile range for passes, median for time) and the two-sided Wilcoxon signed-rank $p$-value for passes (ten seeds). Ratios below one favour ASH-LR. Baselines that did not converge on all seeds are not compared.

| Regime | Baseline | Conv. (base) | Passes ratio | Time ratio | $p$ (passes) |
|---|---|---|---|---|---|
| R3 | Newton-CG | 10/10 | 2.03 [1.87, 2.56] | 2.31 | 0.002 |
| | L-BFGS | 10/10 | 0.71 [0.54, 0.74] | 0.20 | 0.002 |
| | Newton | 10/10 | 5.26 [4.79, 6.10] | 5.25 | 0.002 |
| R4 | Newton-CG | 10/10 | 2.33 [1.99, 2.74] | 2.34 | 0.002 |
| | L-BFGS | 10/10 | 1.80 [1.76, 2.08] | 1.21 | 0.002 |
| | Newton | 10/10 | 1.65 [1.62, 1.92] | 9.51 | 0.002 |
| R5 | Newton-CG | 10/10 | 2.25 [2.23, 2.30] | 2.26 | 0.002 |
| | L-BFGS | 0/10 | baseline not converged | | |
| | Newton | 10/10 | 2.68 [2.68, 2.70] | 16.92 | 0.002 |
| R6 | Newton-CG | 10/10 | 0.93 [0.80, 1.30] | 0.85 | 0.922 |
| | L-BFGS | 0/10 | baseline not converged | | |
| | Newton | 10/10 | 204.40 [142.02, 238.40] | 252.76 | 0.002 |
| R8 | Newton-CG | 10/10 | 1.17 [1.12, 1.31] | 1.28 | 0.014 |
| | L-BFGS | 0/10 | baseline not converged | | |
| | Newton | 10/10 | 4.15 [4.02, 4.45] | 9.02 | 0.002 |

Table 4: E3: ablation of ASH-LR (ten seeds, $w_0 = 0$). Conv. = seeds converged; Passes = median data passes. (a) is the Milestone 2 method and (d) the final method.

| Variant | R2 | | R5 | | R6 | | R8 | |
|---|---|---|---|---|---|---|---|---|
| | Conv. | Passes | Conv. | Passes | Conv. | Passes | Conv. | Passes |
| (a) M2: direct+LS, fixed mu | 10/10 | 118 | 10/10 | 6,817 | 10/10 | 749 | 10/10 | 942 |
| (b) CG+LS, fixed mu | 10/10 | 168 | 10/10 | 6,541 | 10/10 | 48,208 | 10/10 | 2,796 |
| (c) CG+ratio, fixed mu | 10/10 | 168 | 10/10 | 6,541 | 10/10 | 48,208 | 10/10 | 3,033 |
| (d) ASH-LR (full) | 10/10 | 140 | 10/10 | 6,990 | 10/10 | 54,120 | 10/10 | 3,122 |
| (e) full, no GD phase | 10/10 | 91 | 10/10 | 5,686 | 10/10 | 39,316 | 10/10 | 3,192 |
| (f) direct+ratio, grad mu | 10/10 | 105 | 10/10 | 6,817 | 10/10 | 644 | 10/10 | 912 |

*The inner solver dominates in badly scaled problems.* In R6 and R8 the dense-solve variants (a) and (f) needed only 16–21 iterations, whereas every CG variant needed 434–598 iterations in R6 and 67–80 in R8. The cause is the inexact solve: with a Hessian condition number around $10^8$–$10^9$ at the start (Table 1), unpreconditioned CG cannot reduce the residual by the forcing factor within $d$ iterations, so the Newton steps are poor and many of them are needed. Damping cannot repair this: in R8 the ratio test rejected a median of 18 trial steps per run, and the trace in Figure 4 shows $\mu$ jumping to about 20 after a series of rejections before it decays geometrically with the gradient norm.

*Damping rules.* The gradient-regularized rule helped when combined with an exact solve (f vs. a: 11%, 14% and 3% fewer passes in R2, R6 and R8). With CG, it was slightly worse than the fixed rule in R5, R6 and R8 (d vs. c), because a larger $\mu$ early in the Newton phase makes the CG systems easier but the steps shorter. Variants (b) and (c) produced identical iterates in R2, R5 and R6: when every trial step is accepted with $\rho > 0.75$, the ratio test and the line-search rule both halve $\mu$, so the two rules only differ on problems with rejected steps.

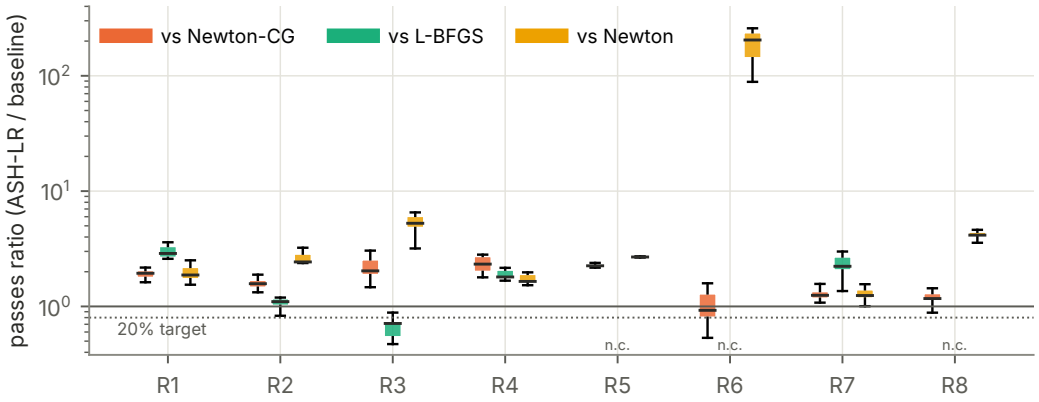



Figure 2: Ratio of data passes, ASH-LR over each baseline, across the ten seeds of every regime (boxes: interquartile range; line: median). Values below the solid line favour ASH-LR; the dotted line marks the 20% target. "n.c.": baseline not converged on all seeds.

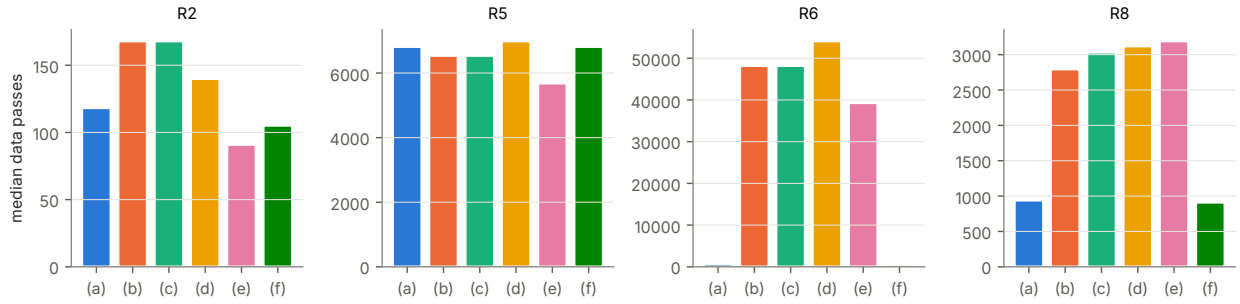



Figure 3: E3: median data passes of the six ASH-LR variants.

### *5.4. Robustness to starting point and regularization (E2)*

Table 5 summarizes 81 problems per method (three regimes, three seeds, three values of $\lambda$, three starting points). ASH-LR, Newton and Newton–CG converged on all 81 problems, never produced a non-finite value and never failed a line search. L-BFGS converged on 47 of 81; it failed on 25 of the 27 raw-Spambase problems and on all nine R5 problems with $\lambda = 10^{-6}$. Robustness is thus shared by all three Newton-type methods and is not a distinguishing advantage of ASH-LR. A distant start hurt ASH-LR most: on R2 its median work grew from 122 passes ($w_0 = 0$) to 1,080 (far start), because with saturated logits the gradient is small and the switching rule keeps the method in the slow gradient phase for longer. Newton–CG grew only from 88 to 132 passes. Larger $\lambda$ improves the conditioning and reduced the work of all methods; for $\lambda = 10^{-2}$ Newton–CG needed a median of 86 passes against 380 for ASH-LR.

### *5.5. Single precision (E4) and numerical stability (E5)*

In single precision with tolerance $10^{-4}$ (Table 6), all three methods converged on R2 and standardized Spambase (R7), with objective gaps comparable to double precision and wall times 4–55% lower. On raw Spambase (R8), however, Newton–CG converged on none and ASH-LR on one of three seeds, while direct Newton converged on all. With a Hessian condition number near $10^9$, which exceeds $1/\epsilon_{\text{float32}} \approx 1.7 \times 10^7$, the rounding errors in the Hessian-vector products and CG recurrences [14] are large enough to make the inexact Newton steps unreliable, whereas the dense

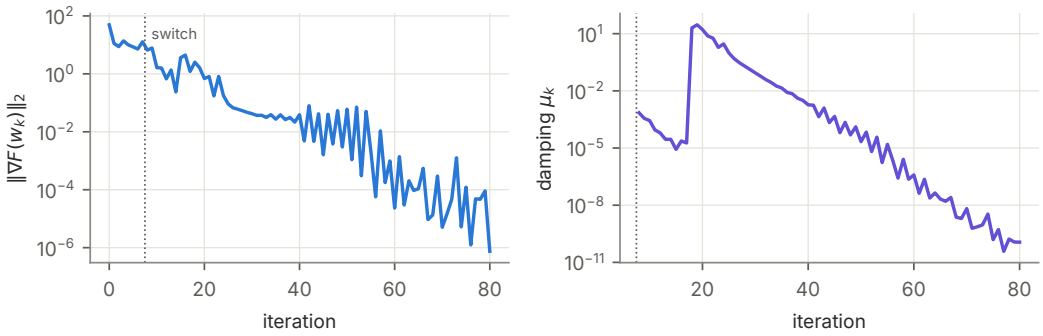



Figure 4: ASH-LR on R8 (seed 1000): gradient norm and damping per iteration. The dotted line marks the switch to the Newton phase.

Table 5: E2: seeds converged and, in parentheses, median data passes over the three seeds and three values of $\lambda \in \{10^{-6}, 10^{-4}, 10^{-2}\}$ (nine problems per cell). Relative stopping rule $\|g\| / \max(1, \|g_0\|) \le 10^{-6}$.

| Regime | Start | Newton | Newton-CG | L-BFGS | ASH-LR |
|---|---|---|---|---|---|
| R2 | zero | 9/9 (50) | 9/9 (88) | 9/9 (111) | 9/9 (122) |
| | near | 9/9 (64) | 9/9 (90) | 9/9 (99) | 9/9 (192) |
| | far | 9/9 (96) | 9/9 (132) | 9/9 (141) | 9/9 (1,080) |
| R5 | zero | 9/9 (2,084) | 9/9 (518) | 6/9 (426) | 9/9 (802) |
| | near | 9/9 (3,127) | 9/9 (532) | 6/9 (470) | 9/9 (1,050) |
| | far | 9/9 (3,662) | 9/9 (686) | 6/9 (530) | 9/9 (1,990) |
| R8 | zero | 9/9 (595) | 9/9 (456) | 0/9 (2,508) | 9/9 (1,040) |
| | near | 9/9 (657) | 9/9 (584) | 1/9 (2,469) | 9/9 (866) |
| | far | 9/9 (921) | 9/9 (471) | 1/9 (2,449) | 9/9 (805) |

factorization still delivers usable steps. Single precision is therefore safe only for reasonably scaled data when the inner solver is CG.

Table 7 reports the stability test from an extreme starting point (logit standard deviation $10^3$). With the stable formulas both methods converged on every problem in double precision. With the naive formulas the initial loss overflows to $+\infty$; every subsequent trial point whose margins still exceed the overflow threshold also evaluates to $+\infty$, so line searches and ratio tests cannot compare candidates. ASH-LR failed in all twelve naive runs, and Newton recovered in only three of the six double-precision runs, when its first full step happened to land at finite values. In single precision, where exp overflows already beyond $|z| \approx 88$, all naive runs failed. The only failures with the stable formulas were the ASH-LR runs on R8 in float32, which reached the iteration limit with finite values, consistent with the CG precision issue above. Stable evaluation is cheap and exact, and it is the one component of ASH-LR whose benefit is unconditional.

### *5.6. Discussion*

The results give a consistent picture of when the ideas behind ASH-LR pay off.

*Stable evaluation* is necessary, not optional: it costs nothing, changes no iterate when the naive formulas are finite, and prevents total failure from distant starting points or in single precision.

*Second-order information* is what makes ill-conditioned logistic regression tractable: all Newton-type methods converged everywhere, while first-order methods and, on badly scaled data, L-BFGS did not. This supports the premise of the project.

Table 6: E4: float64 versus float32 (tolerance $10^{-4}$, three seeds). Median relative gap of the returned solution evaluated in float64, and ratio of median wall times.

| Regime | Method | Conv. | | Median gap | | Time ratio |
|---|---|---|---|---|---|---|
| | | f64 | f32 | f64 | f32 | f32/f64 |
| R2 | Newton | 3/3 | 3/3 | 2.1e-13 | 2.4e-13 | 0.79 |
| | Newton-CG | 3/3 | 3/3 | 1.2e-08 | 1.1e-06 | 0.93 |
| | ASH-LR | 3/3 | 3/3 | 7.8e-07 | 8.9e-07 | 0.96 |
| R7 | Newton | 3/3 | 3/3 | 2.7e-09 | 2.8e-09 | 0.57 |
| | Newton-CG | 3/3 | 3/3 | 1.9e-07 | 2.4e-07 | 0.45 |
| | ASH-LR | 3/3 | 3/3 | 1.1e-05 | 3.2e-07 | 0.53 |
| R8 | Newton | 3/3 | 3/3 | 1.0e-10 | 1.0e-10 | 0.58 |
| | Newton-CG | 3/3 | 0/3 | 1.8e-05 | 5.9e-05 | 29.38 |
| | ASH-LR | 3/3 | 1/3 | 3.3e-05 | 3.0e-05 | 37.60 |

Table 7: E5: seeds converged out of three from an extreme start (logit standard deviation $10^3$) with stable and naive formulas.

| Regime | Precision | Formulas | Newton | ASH-LR | Failure mode |
|---|---|---|---|---|---|
| R2 | float64 | stable | 3/3 | 3/3 | – |
| | float64 | naive | 2/3 | 0/3 | non finite |
| | float32 | stable | 3/3 | 3/3 | – |
| | float32 | naive | 0/3 | 0/3 | non finite |
| R8 | float64 | stable | 3/3 | 3/3 | – |
| | float64 | naive | 1/3 | 0/3 | non finite |
| | float32 | stable | 3/3 | 0/3 | max iter |
| | float32 | naive | 0/3 | 0/3 | non finite |

*The hybrid structure* did not pay off in our regimes. The gradient phase costs 15–40 passes that Newton-type methods do not need from $w_0 = 0$, and it becomes expensive from distant starts. The damping mechanism worked as designed—$\mu$ grew after rejected steps and vanished near the solution—but with CG as the inner solver the bottleneck in badly scaled problems is the accuracy of the inner solve, which damping cannot fix. A diagonal (Jacobi) or randomized Nyström preconditioner [15], or simply feature scaling, would attack the actual cause; the dramatic difference between R7 and R8, which differ only by standardization, makes the same point from the data side.

*Passes versus time.* For $d \leq 500$ dense features, the BLAS-3 Hessian formation of direct Newton runs so much faster per flop than the matrix–vector products of CG that direct Newton wins in time even where it loses in passes (R4). Matrix-free methods become necessary only when $d$ is so large that $d^2$ memory or $d^3$ work is prohibitive, or when $X$ is sparse; our experiments do not cover that regime, so they cannot show whether ASH-LR would then compete with Newton–CG.

*Threats to validity.* Timings come from one virtual machine with two cores and three repetitions; passes, which are machine-independent, lead to the same conclusions. The synthetic designs have log-spaced spectra and logistic labels and may be easier or harder than real data. Spambase contains 391 duplicate rows, which affects test accuracy but not the optimization comparison. Finally, the ASH-LR parameters (switching thresholds, $\mu_0$, update factors) were not tuned. The ablation bounds what tuning the switch could achieve: even with no gradient phase at all (variant e), ASH-LR matched Newton–CG in R2 (91 vs. 92 passes), used 15% fewer passes in R6 (39,316 vs. 46,162), and stayed behind in R5 and R8. This post-hoc observation suggests that a damped Newton–CG without the gradient phase may be competitive on badly scaled problems, but it would need its own

pre-registered test.

## 6. Conclusions

We implemented and evaluated ASH-LR, a hybrid method for ill-conditioned $\ell_2$-regularized logistic regression that combines stable loss evaluation, an initial gradient phase, and a damped inexact Newton–CG phase with a ratio test and gradient-regularized damping $\mu_k = \sqrt{\theta_k \|g_k\|}$. The evaluation followed a plan fixed before the final experiments, with eight regimes, ten seeds, five baselines and four supplementary experiments.

ASH-LR converged on every problem in double precision, including those where gradient descent and L-BFGS failed, and stable evaluation was essential for convergence from distant starting points. However, the pre-registered efficiency hypothesis was not supported: in none of the five primary ill-conditioned regimes did ASH-LR reduce the work of the best matrix-free baseline, Newton–CG, by 20%; it required 18% to 127% more data passes in four regimes and was statistically on a par in one badly scaled regime. Direct Newton was the fastest method throughout the dimensions studied ($d \leq 500$). The ablation located the causes: the gradient phase adds work that Newton-type methods do not need from $w_0 = 0$, and in badly scaled problems the accuracy of the unpreconditioned CG solve, not the damping, limits progress.

For practitioners, the results suggest a simple rule: evaluate the loss stably, use direct Newton when $d$ is small enough to form the Hessian, and otherwise use Newton–CG, preferably on standardized features. Future work should (i) add a diagonal or randomized Nyström preconditioner [15] to the CG solver, which targets the observed bottleneck; (ii) test damped Newton–CG without the gradient phase, which the ablation identified as promising on badly scaled data, in a new pre-registered study; (iii) extend the evaluation to high-dimensional sparse data, where matrix-free methods are the only option; and (iv) analyse the convergence of the gradient-regularized ratio-test rule with inexact solves.

## Code and data availability

The implementation, tests, experiment scripts, raw results and the pre-registered plan are available in the project repository (see the README for the URL). Spambase is available from the UCI Machine Learning Repository under CC BY 4.0 [18].

## References


[1] S. J. Wright, B. Recht, Optimization for Data Analysis, Cambridge University Press, Cambridge, 2022.

[2] A. S. Berahas, R. Bollapragada, J. Nocedal, An investigation of Newton-Sketch and subsampled Newton methods, Optimization Methods and Software 35 (4) (2020) 661–680. doi:10.1080/10556788.2020.1725751.

[3] P. Blanchard, D. J. Higham, N. J. Higham, Accurately computing the log-sum-exp and softmax functions, IMA Journal of Numerical Analysis 41 (4) (2021) 2311–2330. doi:10.1093/imanum/draa038.

[4] P. F. V. Wiemann, T. Kneib, J. Hambuckers, Using the softplus function to construct alternative link functions in generalized linear models and beyond, Statistical Papers 65 (2024) 3155–3180. doi:10.1007/s00362-023-01509-x.

[5] K. Mishchenko, Regularized Newton method with global $O(1/k^2)$ convergence, SIAM Journal on Optimization 33 (3) (2023) 1440–1462.

[6] N. Doikov, K. Mishchenko, Y. Nesterov, Super-universal regularized Newton method, SIAM Journal on Optimization 34 (1) (2024) 27–56. doi:10.1137/22M1519444.

[7] L. Galli, C.-J. Lin, A study on truncated Newton methods for linear classification, IEEE Transactions on Neural Networks and Learning Systems 33 (2022) 2828–2841. doi:10.1109/TNNLS.2020.3045836.

[8] A. S. Berahas, M. Jahani, P. Richtárik, M. Takáč, Quasi-Newton methods for machine learning: forget the past, just sample, Optimization Methods and Software 37 (5) (2022) 1668–1704. doi:10.1080/10556788.2021.1977806.

[9] C. W. Royer, M. O'Neill, S. J. Wright, A Newton-CG algorithm with complexity guarantees for smooth unconstrained optimization, Mathematical Programming 180 (2020) 451–488. doi:10.1007/s10107-019-01362-7.

[10] Z. Yao, P. Xu, F. Roosta, S. J. Wright, M. W. Mahoney, Inexact Newton-CG algorithms with complexity guarantees, IMA Journal of Numerical Analysis 43 (3) (2023) 1855–1897. doi:10.1093/imanum/drac043.

[11] A. Lim, F. Roosta, Complexity guarantees for nonconvex Newton-MR under inexact Hessian information, IMA Journal of Numerical Analysis 46 (1) (2026) 372–399. doi:10.1093/imanum/drae110.

[12] N. Doikov, Y. Nesterov, Gradient regularization of Newton method with Bregman distances, Mathematical Programming 204 (1–2) (2024) 1–25. doi:10.1007/s10107-023-01943-7.

[13] S. Hanzely, D. Kamzolov, D. Pasechnyuk, A. Gasnikov, P. Richtárik, M. Takáč, A damped Newton method achieves global $O(1/k^2)$ and local quadratic convergence rate, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022), 2022.

[14] N. J. Higham, T. Mary, Mixed precision algorithms in numerical linear algebra, Acta Numerica 31 (2022) 347–414. doi:10.1017/S0962492922000022.

[15] Z. Frangella, J. A. Tropp, M. Udell, Randomized Nyström preconditioning, SIAM Journal on Matrix Analysis and Applications 44 (2) (2023) 718–752. doi:10.1137/21M1466244.

[16] A. Godichon-Baggioni, N. Werge, On adaptive stochastic optimization for streaming data: A Newton's method with $O(dN)$ operations, Journal of Machine Learning Research 26 (59) (2025) 1–49.

[17] T. Moreau, M. Massias, A. Gramfort, P. Ablin, P.-A. Bannier, et al., Benchopt: Reproducible, efficient and collaborative optimization benchmarks, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022), 2022.

[18] M. Hopkins, E. Reeber, G. Forman, J. Suermondt, Spambase, UCI Machine Learning Repository, license CC BY 4.0 (1999). doi:10.24432/C53G6X.